

Software corp

Software Requirement Specification

for

ParkPilot (Smart Parking Platform)

Version 1.0

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Software Requirements Specification

ParkPilot

Version 1.0

1. Existing Software Landscape

1.1 Existing Parking Applications:

This purpose of this document will be to research existing parking services that are similar to ParkPilot. SpotHero will be used as the primary competitor, along with one or two additional parking platforms for comparison. The research will examine the features offered by these applications and identify ideas that may influence the development of ParkPilot.

1.2 ParkPilot Differentiation:

This section will explain how ParkPilot will differ from existing parking applications. Rather than simply duplicating an existing platform, ParkPilot will identify opportunities to improve the parking experience for drivers and parking operators, with a focus on the Houston area. ParkPilot will differentiate itself from existing parking applications by providing a more complete and locally focused parking experience for the Houston area. The platform will combine real-time parking availability, reservations, digital payments, navigation, and parking facility information on one application. Users will be able to compare locations based on price, distance, availability, and amenities, while features such as QR-code or license-plate detection, parking reminders, and reservation extensions will make parking more convenient. ParkPilot will also provide parking operators with tools to monitor occupancy, pricing, reservations, and facility usage.

2. Vision and Scope

2.1 Project Acquisition:

The ParkPilot project was acquired by **Software ¢orp.** after identifying a need for a more efficient parking solution in the Houston metropolitan area. Drivers frequently experience difficulty locating available and affordable parking, particularly in high-traffic areas. Software ¢orp. will develop ParkPilot to provide drivers and parking operators with a centralized platform for locating, reserving, managing, and paying for parking.

2.2 Company Introduction:

Software corp. is a software development company that specializes in creating technology solutions designed to improve everyday business operations. The company develops web and mobile applications with a focus on usability, reliability, and efficiency. For the ParkPilot project, Software corp. will oversee the planning, development, testing, and implementation of the smart parking platform.

2.3 Project Overview:

ParkPilot is a smart parking platform designed to help drivers locate available parking spaces, compare parking options, reserve spaces, make digital payments, and navigate participating parking facilities. The platform will be available through both web and mobile applications and will initially focus on the Houston metropolitan area. Parking operators will also have access to management tools for monitoring occupancy, reservations, pricing, and parking activity.

3. Software Requirements Specification (SRS)

3.1 Use Cases

Use Case 1

Title: Register Driver Account

Actor: Driver

Precondition: User does not already have a ParkPilot account.

Steps:

1. User selects "Create Account."
2. User enters their name, email address, phone number, and password.
3. User reviews the entered information.
4. User selects "Create Account."
5. ParkPilot creates the new driver account.

Postcondition: Driver account is created and available for use.

Use Case 2

Title: Log In and Authenticate User

Actor: Driver or Parking Operator

Precondition: User has a registered ParkPilot account.

Steps:

1. User selects “Log In.”
2. User enters their email address and password.
3. User submits the login information.
4. ParkPilot verifies the user’s credentials.
5. User is granted access to the appropriate account.

Postcondition: User is authenticated and logged into ParkPilot.

Use Case 3

Title: Register Vehicle

Actor: Driver

Precondition: Driver is logged into ParkPilot.

Steps:

1. Driver selects “My Vehicles.”
2. Driver selects “Add Vehicle.”
3. Driver enters the vehicle license plate, make, model, and color.
4. Driver reviews the vehicle information.
5. Driver selects “Save Vehicle.”

Postcondition: Vehicle is registered to the driver’s account.

Use Case 4

Title: Register Payment Method

Actor: Driver

Precondition: Driver is logged into ParkPilot.

Steps:

1. Driver selects “Payment Methods.”
2. Driver selects “Add Payment Method.”
3. Driver enters the required payment information.
4. ParkPilot validates the payment information.
5. Driver saves the payment method.

Postcondition: Payment method is securely associated with the driver’s account.

Use Case 5

Title: Find Parking Garage

Actor: Driver

Precondition: Driver has access to the ParkPilot application.

Steps:

1. Driver enters a destination or location.
2. Driver enters the desired parking date and time.
3. ParkPilot searches participating parking facilities.
4. ParkPilot displays available garages on an interactive map.
5. Driver selects a garage to view additional information.

Postcondition: Driver is shown available parking options near the selected destination.

Use Case 6

Title: Reserve Parking Space

Actor: Driver

Precondition: Driver is logged in and has selected an available parking facility.

Steps:

1. Driver selects the desired garage.
2. Driver selects the parking date and time.
3. Driver selects a registered vehicle.

4. ParkPilot displays the reservation details and price.
5. Driver confirms the reservation.

Postcondition: A parking space is reserved for the driver.

Use Case 7

Title: Pay for Parking Reservation

Actor: Driver

Precondition: Driver has selected a parking reservation and has a valid payment method.

Steps:

1. Driver reviews the reservation cost.
2. Driver selects a registered payment method.
3. Driver selects “Pay.”
4. ParkPilot processes the payment.
5. ParkPilot displays a payment confirmation.

Postcondition: Parking reservation is paid and confirmed.

Use Case 8

Title: Navigate to Parking Garage

Actor: Driver

Precondition: Driver has selected or reserved a parking facility.

Steps:

1. Driver selects “Directions.”
2. ParkPilot sends the garage location to the navigation service.
3. Navigation directions are displayed.
4. Driver follows the directions to the parking facility.

Postcondition: Driver receives navigation assistance to the selected garage.

Use Case 9

Title: Check In to Parking Facility

Actor: Driver

Precondition: Driver has an active reservation at the parking facility.

Steps:

1. Driver arrives at the reserved parking facility.
2. Driver scans the ParkPilot QR code or uses the registered license plate.
3. ParkPilot verifies the active reservation.
4. The parking facility grants the driver access.
5. ParkPilot records the driver's arrival.

Postcondition: Driver is successfully checked into the parking facility.

Use Case 10

Title: Extend Parking Time

Actor: Driver

Precondition: Driver has an active parking reservation and additional time is available.

Steps:

1. Driver opens the active reservation.
2. Driver selects "Extend Parking."
3. Driver selects a new ending time.
4. ParkPilot displays any additional cost.
5. Driver confirms and pays for the extension.

Postcondition: Parking reservation is extended to the new ending time.

Use Case 11

Title: Cancel Parking Reservation

Actor: Driver

Precondition: Driver has an active reservation that is eligible for cancellation.

Steps:

1. Driver opens the reservation.
2. Driver selects “Cancel Reservation.”
3. ParkPilot displays the cancellation information.
4. Driver confirms the cancellation.
5. ParkPilot cancels the reservation and processes any applicable refund.

Postcondition: Reservation is canceled and the parking space becomes available again.

Use Case 12

Title: View Reservation History and Receipt

Actor: Driver

Precondition: Driver is logged into ParkPilot and has completed at least one reservation.

Steps:

1. Driver selects “Reservation History.”
2. ParkPilot displays current and previous reservations.
3. Driver selects a reservation.
4. ParkPilot displays the reservation details and payment information.
5. Driver views or downloads the receipt.

Postcondition: Driver can review previous parking reservations and receipts.

Use Case 13

Title: Register Parking Operator Account

Actor: Parking Operator

Precondition: Parking operator does not already have a ParkPilot operator account.

Steps:

1. Operator selects “Register as Parking Operator.”
2. Operator enters business and contact information.

3. Operator creates login credentials.
4. Operator submits the registration.
5. ParkPilot creates the operator account.

Postcondition: Parking operator account is created and can be used to manage parking facilities.

Use Case 14

Title: Manage Parking Facility

Actor: Parking Operator

Precondition: Operator is logged into an approved ParkPilot operator account.

Steps:

1. Operator opens the management dashboard.
2. Operator selects a parking facility.
3. Operator updates parking capacity, availability, hours, amenities, or pricing.
4. Operator reviews the changes.
5. Operator saves the updated facility information.

Postcondition: Updated parking facility information is available to ParkPilot users.

Use Case 15

Title: View Parking Occupancy and Analytics

Actor: Parking Operator

Precondition: Operator is logged in and manages at least one parking facility.

Steps:

1. Operator opens the management dashboard.
2. Operator selects “Reports and Analytics.”
3. Operator selects a parking facility and reporting period.
4. ParkPilot displays occupancy, reservations, revenue, and usage information.
5. Operator reviews the report.

Postcondition: Parking operator can review facility performance and parking activity.